

Investigating a Self-Starting Siphon

Qihang Lu | Shanghai Pinghe School

This poster presents my investigation of how a specially shaped rigid tube can start siphoning water without initial suction or pre-filling. Focusing on an M-shaped acrylic tube, the project explores how geometry affects both successful self-starting and the speed of water emerging from the outlet.

Research question

How do the height of the highest bend above the water surface, tube segment lengths, internal radius, and outlet angle influence the siphoning process? The central challenge is understanding whether the moving water gains enough kinetic energy to pass the upper bend and establish continuous flow.

Approach

The investigation combines preliminary observations, a theoretical model, and repeated experiments. The model divides the tube into four straight sections and three bends, accounting for pressure, gravity, capillary forces, and flow resistance. Experiments use acrylic siphon tubes, a beaker, and a support stand. Changing the water level and tube geometry allows comparisons between configurations, while video tracking provides estimates of outlet velocity.

Main findings

Within the tested conditions, increasing the height of the highest bend above the water surface reduced outlet velocity. A longer second rising section produced higher outlet speeds in the compared configurations. Increasing the internal radius from 2.0 to 3.0 mm raised measured outlet velocity from 13.9 to 17.6 cm/s; however, larger tubes encountered trapped air that prevented successful siphoning. Increasing the angle between the final two sections also reduced outlet velocity in the tested range.

Significance and limitations

The project connects fluid mechanics with a practical design question: how can geometry enable a device to start operating on its own? It also demonstrates the importance of testing theoretical predictions against physical observations. Video resolution limited the detection of smaller effects, and the simplified model did not account for variations in velocity across the tube or the detailed shape of the water-air interface.